

Bilinear ergodic Hilbert variation for commuting transformations as a consequence of annular triangular variation

Alec Kriebel

ORCID: 0009-0001-9320-500X

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Abstract

We give an explicit restriction and transference consequence of OpenAI's October 5, 2026 full annular variation theorem for the triangular Hilbert transform. For arbitrary commuting invertible probability-preserving transformations S, T and complex $f, g \in L^3$, the symmetric bilinear Hilbert sums have bounded pointwise r -variation in $L^{3/2}$ for every $r > 2$, and converge almost everywhere and in $L^{3/2}$. The restriction uses boxes of fixed positive area and a uniformly summable kernel error. The analytic breakthrough is the cited continuous theorem; the present note supplies the discrete restriction and full measure-theoretic transfer.

1 Statement and provenance

Let (X, Σ, μ) be a probability space. Assume that S, T are invertible measurable measure-preserving transformations with measurable inverses, and $ST = TS$. For complex $f, g \in L^3(X)$ put

$$H_0(f, g) = 0, \quad H_N(f, g)(x) = \sum_{0 < |n| \leq N} \frac{f(S^n x)g(T^n x)}{n} \quad (N \geq 1).$$

For a complex sequence $(a_N)_{N \geq 0}$ define the homogeneous variation

$$V^r(a_N : N \geq 0) = \sup_{\substack{J \geq 1 \\ 0 \leq N_0 < \dots < N_J, \ N_j \in \mathbb{Z}}} \left(\sum_{j=1}^J |a_{N_j} - a_{N_{j-1}}|^r \right)^{1/r}. \quad (1)$$

If $a_0 = 0$, prepending 0 to any list shows that restricting the supremum to $N_0 = 0$ gives the same value. In particular $\sup_N |a_N| \leq V^r(a)$.

Theorem 1. *For every $r > 2$ there is a constant $D_r < \infty$, independent of the probability space and transformations, such that*

$$\|V^r(H_N(f, g) : N \geq 0)\|_{3/2} \leq D_r \|f\|_3 \|g\|_3. \quad (2)$$

The maximal function satisfies the same bound with V^r replaced by $\sup_{N \geq 0} |H_N|$. The sums converge almost everywhere and in $L^{3/2}$ to a complex bilinear operator $H(f, g)$, and moreover

$$\left\| \sup_{N \geq m} |H_N(f, g) - H(f, g)| \right\|_{3/2} \longrightarrow 0. \quad (3)$$

The single-transformation Hilbert theorem of Demeter [3, Theorem 1.2] treats opposite powers of one transformation. Becker and Durcik [1, Corollary 1.4] prove sharp $r > 2$ variation for bilinear averages along opposite powers of one transformation; their discussion of equation (1.8) distinguishes the arbitrary commuting case. Demeter and Thiele [4, Section 6] relate the triangular transform to two commuting transformations. Calderón’s transference method [2] is established machinery; we include its finite-window argument rather than appeal to it without quantifiers. The continuous-to-lattice and orbit-array reductions already occur in [4, Section 6, footnote 36]; positive-area phase integration also appears in [5, Section 5]. The fixed-width singular-kernel bookkeeping below makes this particular consequence explicit.

The new input is the following continuous estimate, stated as Theorem 1.1 in the October 5 annular manuscript [6]. For $F, G \in L^3(\mathbb{R}^2)$ write

$$B_{a,b}(F, G)(u, v) = \int_{a < |t| < b} F(u + t, v) G(u, v + t) \frac{dt}{t}.$$

Then, for every $r > 2$,

$$\left\| \sup_{\substack{J \geq 1 \\ 0 < t_0 < \dots < t_J, \, t_j \in \mathbb{Q}}} \left(\sum_{j=1}^J |B_{t_{j-1}, t_j}(F, G)|^r \right)^{1/r} \right\|_{L^{3/2}(\mathbb{R}^2)} \leq C_r \|F\|_3 \|G\|_3. \quad (4)$$

The supremum is inside the norm and partitions may depend on (u, v) . It includes all positive scales, with unrestricted finite partition length. We use precisely this estimate, rather than a maximal, dyadic, or fixed-partition norm-variation estimate. The upstream theorem is not reproved here; all conclusions below are consequences of it. The listed family 082 Lean scope concerns an older maximal theorem and does not formally verify (4) or this note.

Our contribution is the fixed-width restriction proof in Section 2 and its explicit transfer in Section 3. We do not claim independent resolution of the continuous triangular Hilbert problem or invention of transference. Manuscript dates in the upstream collection are not assertions of first public disclosure. The source repository version used is

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2 Fixed-width restriction to the lattice

For finitely supported arrays $A, B : \mathbb{Z}^2 \rightarrow \mathbb{C}$ put

$$h_N(A, B)(i, j) = \sum_{0 < |k| \leq N} \frac{A(i + k, j) B(i, j + k)}{k}, \quad h_0 = 0.$$

All lattice norms use counting measure.

Proposition 2. *Assuming (4),*

$$\|V^r(h_N : N \geq 0)\|_{\ell^{3/2}(\mathbb{Z}^2)} \leq D_r \|A\|_{\ell^3} \|B\|_{\ell^3}, \quad D_r = 8 \cdot 4^{2/3} C_r + \frac{\pi^2}{13}. \quad (5)$$

Proof. Fix $h = 1/8$ and $q = 1/16$. Extend A, B to F, G on $\mathbb{R}^2$ by making them constant with their corresponding array values on

$$(i, j) + [-h, h]^2 \quad ((i, j) \in \mathbb{Z}^2)$$

and zero elsewhere. Boundary values have no effect and can be assigned arbitrarily. The boxes are disjoint and

$$\|F\|_3 = (4h^2)^{1/3} \|A\|_{\ell^3}, \quad \|G\|_3 = (4h^2)^{1/3} \|B\|_{\ell^3}. \quad (6)$$

Evaluate at $(u, v) = (i + a, j + b)$ with $a, b \in (-q, q)$. If $F(u + t, v)G(u, v + t)$ is nonzero, the fixed coordinate v selects the second cell index j of F , and the fixed coordinate u selects the first cell index i of G . The moving coordinates require

$$t \in k + [-h - a, h - a] \quad \text{and} \quad t \in l + [-h - b, h - b]$$

for integers k, l . If these intervals overlap, then $|k - l| \leq 2h + |a - b| \leq 3/8 < 1$, so $k = l$. Consequently the integrand equals the single lattice product $A(i + k, j)B(i, j + k)$ on $k + I_{a,b}$, where

$$I_{a,b} = [-h - \min(a, b), h - \max(a, b)], \quad \ell(a, b) = |I_{a,b}| = 2h - |a - b| \geq 1/8. \quad (7)$$

Also $\ell \leq 1/4$ and every $s \in I_{a,b}$ satisfies $|s| \leq 3/16$. Different integer windows are disjoint.

Set $\rho_N = N + 1/2$ for $N \geq 0$. For $k \neq 0$, the entire window $k + I_{a,b}$ lies in $|t| \in [|k| - 3/16, |k| + 3/16]$. Thus $\rho_M < |t| < \rho_N$ includes that window exactly when $M < |k| \leq N$. This verifies negative as well as positive windows. The window at $k = 0$ lies in $|t| \leq 3/16$ and is excluded even at the smallest endpoint $\rho_0 = 1/2$. Hence for $0 \leq M < N$,

$$\begin{aligned} B_{\rho_M, \rho_N}(F, G)(i + a, j + b) &= \sum_{M < |k| \leq N} A(i + k, j)B(i, j + k)K_k(a, b), \\ K_k(a, b) &= \int_{I_{a,b}} \frac{ds}{k + s}. \end{aligned} \quad (8)$$

There is no principal value in this identity. In particular $k + s$ has the sign of k .

Writing $K_k = \ell/k + \delta_k$, the exact identity $1/(k + s) - 1/k = -s/[k(k + s)]$ gives the uniform bound

$$\frac{|\delta_k(a, b)|}{\ell(a, b)} \leq \frac{3}{13} \frac{1}{k^2} \quad (k \neq 0), \quad (9)$$

since $|k + s| \geq (13/16)|k|$. Let $E_N = \sum_{0 < |k| \leq N} \delta_k A(i + k, j)B(i, j + k)$ and $E_0 = 0$. For any increasing integer list, the radial shells of its increments are disjoint. The ℓ^r norm of the error increments is at most their ℓ^1 norm, which is at most the sum of all absolute error terms. Therefore

$$\ell^{-1} V^r(E_N : N \geq 0) \leq e(i, j) := \frac{3}{13} \sum_{k \neq 0} \frac{|A(i + k, j)B(i, j + k)|}{k^2}.$$

Minkowski, Hölder, and translation invariance imply

$$\|e\|_{\ell^{3/2}} \leq \frac{3}{13} \sum_{k \neq 0} \frac{1}{k^2} \|A\|_{\ell^3} \|B\|_{\ell^3} = \frac{\pi^2}{13} \|A\|_{\ell^3} \|B\|_{\ell^3}. \quad (10)$$

This bounds the whole error variation, not merely each truncation.

Let $W(F, G)$ denote the pointwise supremum in (4). The half-integer endpoints are rational. Taking the supremum over integer lists in (8), using the variation triangle inequality and $\ell^{-1} \leq 8$, gives

$$V^r(h_N)(i, j) \leq 8W(F, G)(i + a, j + b) + e(i, j) \quad (11)$$

for almost every $(a, b) \in (-q, q)^2$, for every (i, j) . All lists are countable, so one common exceptional set suffices. Apply the $L^{3/2}$ triangle inequality on the disjoint output boxes, whose area is $4q^2 = 1/64$. The left side is constant on each box, so (11), (4) and (6) give

$$\|V^r(h_N)\|_{\ell^{3/2}} \leq 8(4q^2)^{-2/3} C_r (4h^2)^{2/3} \|A\|_{\ell^3} \|B\|_{\ell^3} + \|e\|_{\ell^{3/2}}.$$

The coefficient equals $8 \cdot 4^{2/3} C_r$, proving the proposition. The restriction is carried by a fixed set of positive area; no evaluation on a null set of suspension phases is used. $\square$

3 Finite-window transference and convergence

Proof of Theorem 1. Choose measurable representatives of f, g taking finite values outside their null sets. Remove all preimages of those null sets under finite words in S, S^{-1}, T, T^{-1} . The resulting invariant full-measure set carries every value in the finite sums. A change of representatives affects only such a countable union of null sets. The same convention treats transformations commuting modulo null sets: remove all such word preimages of the failure set, after which all required identities hold on the invariant complement.

Fix an integer $M \geq 1$. Let $W_M(x)$ be the maximum in (1) over lists with endpoints in $\{0, \dots, M\}$. There are finitely many such lists, so W_M is measurable. For $L \geq 1$ let $Q_L = \{-L, \dots, L\}^2$. At each x form finite arrays

$$\begin{aligned} A_x(i, j) &= f(S^i T^j x) \mathbf{1}_{Q_{L+M}}(i, j), \\ B_x(i, j) &= g(S^i T^j x) \mathbf{1}_{Q_{L+M}}(i, j). \end{aligned}$$

For $(i, j) \in Q_L$ and $|n| \leq M$, every required entry is in Q_{L+M} . Commutation gives exactly

$$h_N(A_x, B_x)(i, j) = H_N(f, g)(S^i T^j x) \quad (0 \leq N \leq M).$$

Proposition 2, with $p = 3/2$, implies

$$\sum_{(i,j) \in Q_L} W_M(S^i T^j x)^p \leq D_r^p \left(\sum_{Q_{L+M}} |f(S^i T^j x)|^3 \right)^{1/2} \left(\sum_{Q_{L+M}} |g(S^i T^j x)|^3 \right)^{1/2}.$$

Integrating, measure invariance on the left and Cauchy–Schwarz on the right give

$$|Q_L| \|W_M\|_p^p \leq D_r^p |Q_{L+M}| \|f\|_3^{3/2} \|g\|_3^{3/2}.$$

The finite right side establishes integrability without assuming it in advance. For fixed M , let $L \rightarrow \infty$; $|Q_{L+M}|/|Q_L| \rightarrow 1$. Finally W_M increases to the full countable supremum (1). Monotone convergence proves (2). Partitions may depend on the output throughout.

Fix, for example, $r = 3$. The variation is finite almost everywhere. Any scalar sequence with finite r -variation is Cauchy: otherwise some $\varepsilon > 0$ admits arbitrarily late pairs of indices whose values differ by at least ε . Recursively choose J strictly ordered disjoint pairs. A partition containing their endpoints has variation at least $\varepsilon J^{1/r}$, a contradiction. Completeness of $\mathbb{C}$ gives a finite limit H . It is measurable as the limit along integers, setting it to zero on the exceptional set.

Since $H_0 = 0$, $H_* := \sup_N |H_N| \leq V^3(H_N)$ belongs to L^p . Also $|H| \leq H_*$. The measurable tail $\sup_{N \geq m} |H_N - H|$ tends to zero pointwise and is bounded by $2H_*$. Dominated convergence applied to its p th power proves (3), hence norm convergence. The maximal bound follows from (2); the limit bound follows from domination. Norm limits of the bilinear identities for H_N show that H is complex bilinear on $L^3 \times L^3$. $\square$

4 Scope and verification disclosure

The truncations are symmetric and the kernel is odd. This proof gives no one-sided commuting Cesàro convergence, no noncommuting theorem, no $r = 2$ estimate, and no extension of the input exponents beyond L^3 . No numerical evidence substitutes for the analytic estimate. Supplemental exact kernel checks test only Section 2 bookkeeping.

The continuous theorem is attributed to OpenAI using its supplied manuscript citation. The dependency ledger records the version and scope of independent automated proof audits. Such audits are evidence, not formal certification. No theorem in this note is claimed to have been formalized. AI tools were used extensively in research, drafting and verification. This preprint has not undergone conventional human peer review or refereeing at publication.

References

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